

Speaker Extraction for Multi-User Studies or Why Clip-on Mics Are Not Enough

Adrien Coppens
Luxembourg Institute of Science and
Technology
Esch-sur-Alzette, Luxembourg
adrien.coppens@list.lu

Johannes Hermen
Luxembourg Institute of Science and
Technology
Esch-sur-Alzette, Luxembourg
johannes.hermen@list.lu

Valérie Maquil
Luxembourg Institute of Science and
Technology
Esch-sur-Alzette, Luxembourg
valerie.maquil@list.lu

Abstract

Speech is a common modality for interactive systems and collaboration analysis in multi-user studies. Despite the widespread availability of automated transcription tools, attributing speech reliably to individual participants remains challenging in collocated settings. Yet, microphone type and placement are often overlooked, considered as pragmatic constraints rather than being treated as design variables. In this paper, we present a controlled study evaluating how wearable microphone configurations affect the robustness of target speaker extraction using a lightweight speech detection approach. Three configurations were compared: a shirt-mounted clip-on microphone, the same microphone positioned near the wearer's mouth via a rigid necklace, and a commercial headset. We show that shirt-mounted clip-ons often fail to isolate the wearer's speech from the non-wearer's, whereas near-mouth and headset placements substantially improve robustness. These findings highlight microphone choice as a key design decision for speech-based interactive systems and multi-user studies, providing empirically grounded guidance for researchers and practitioners.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; **Collaborative and social computing systems and tools**; • **Hardware** → *Sound-based input / output*.

Keywords

Speaker extraction, Collaboration, Microphone, Speech-based Interaction, Conversational analysis

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1 Introduction

Speech is widely used in Human-Computer Interaction (HCI) both as direct input and as a source of data to study human activity. In addition to being used as an input channel, speech is also frequently analyzed in multi-user studies and collaborative systems

to support qualitative and quantitative analyses of collaboration, communication patterns, and interaction dynamics [1], as verbal communication is central for coordination and sense-making [5].

Recent advances in automatic speech recognition (ASR) make AI-based transcription services widely available to researchers, and allow the processing of large amounts of audio data with relatively low effort. Open-weight speech-to-text models such as Whisper [8] as well as many commercial services now open the door to automated transcriptions and analysis of speech data in user studies and research settings [4]. This creates new opportunities for studying verbal interaction at scale. At the same time, these advances highlight a persistent challenge: reliably attributing speech to individual speakers. Speaker diarization remains a difficult problem, especially when combined with ASR and in collocated settings where multiple people speak in close proximity and overlap in time [7].

Despite the central role that microphone input plays in speech-based analysis and interactive systems, microphone choice and placement are often treated as pragmatic decisions rather than subjected to systematic evaluation. Researchers therefore lack empirical guidance on how different wearable microphone configurations affect the reliability of speaker extraction, particularly when using simple processing pipelines rather than sophisticated machine learning models.

In this paper, we aim to answer the following question: “Is a clip-on microphone sufficient to isolate a wearer's speech from nearby speakers using simple speech detection?” (RQ1). Then, we follow-up with “How does recording equipment and microphone placement affect speaker extraction robustness in realistic multi-user study settings?” (RQ2).

To answer these questions, we present a controlled, practically-grounded evaluation of wearable microphone alternatives for target speaker extraction using a simple Voice Activity Detection (VAD) pipeline. We compare three configurations: a shirt-mounted clip-on microphone, the same microphone positioned closer to the mouth using a rigid necklace, and a commercial headset microphone. Using a structured recording protocol involving different speaker distances and voice levels, we examine how microphone placement influences the ability to distinguish the speech of the wearer from that of the non-wearer. Our findings show that clip-on microphones attached to clothing are often insufficient for reliable target speaker extraction, while near-mouth and headset microphones substantially improve the robustness of the approach. We conclude by discussing the implications of the results for designing speech-based interactive systems and study setups.



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2 Related Work

One common approach to capture speech in shared spaces relies on room microphones. While such setups are convenient and minimally intrusive, they make the problem of speaker attribution prominent, as all voices are then captured through a single device. In collocated and multi-user settings, multiple participants typically speak in close proximity or overlap. Accurately separating speakers from such room microphone recordings is therefore challenging and requires complex processing pipelines and powerful models. This often leads researchers to opt for manual post-processing steps after obtaining transcriptions results [6].

As an alternative, many have turned to individual and wearable microphones to obtain a signal that is more closely associated with a particular user. In practice, this often involves simple microphone configurations such as clip-on microphones attached to clothing, as they can capture speech close to a participant's mouth, in a relatively unobtrusive manner. Such devices are generally easy to deploy, comfortable and socially accepted. Their proximity to the mouth is expected to enable lightweight speech detection techniques (typically referred to as Voice Activity Detection (VAD) [10]) to successfully differentiate the speech of the wearer from that of other collocated persons. Such approaches are commonly used in laboratory studies because they are inexpensive and compatible even with real-time interaction. However, it remains unclear to what extent common wearable microphone placements are sufficient to isolate the wearer's speech when other people are nearby in shared spaces. Despite their widespread use, microphone placement is rarely treated as an explicit variable of interest. The few existing comparative evaluations (e.g. [3]) focus on recognition rates rather than investigating performance differences in speaker detection.

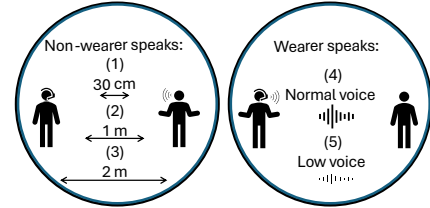
In many scenarios, lightweight VAD tools can determine when a user is speaking while remaining computationally inexpensive, introducing minimal latency, which make them well suited for real-time interaction [9]. They can then be used as building blocks for higher-level automated decisions, such as turn-taking detection, attention estimation, or conversational triggers. While substantial research in speech processing has focused on improving VAD accuracy using advanced machine learning models [11], such approaches may be impractical for interactive systems that require low latency, robustness, and ease of deployment. In this context, the effectiveness of lightweight speech detection depends not only on the algorithm itself but also on the quality and characteristics of the audio input. However, the interaction between microphone placement and simple speech detection pipelines remains underexplored in HCI research.

In contrast to prior work that advances speech processing techniques or assumes a fixed audio capture setup, our work focuses on wearable microphone alternatives and on the impact of placement for target speaker extraction. We present a controlled, comparative evaluation of common wearable microphone configurations using a simple VAD pipeline, with the goal of providing practical guidance for designers of speech-based interactive systems.

3 Study Design and Data Collection

The goal of our study was to isolate the effect of wearable microphone placement on the ability to distinguish a wearer's speech

from nearby speakers using a lightweight processing pipeline. Rather than modeling natural conversation, we adopted a controlled recording protocol that allowed us to systematically vary speaker distance, voice level and microphone placement while keeping all other factors constant. All recordings were conducted in an indoor room we commonly use for studies on interactive (multi-user) systems in our lab. They followed the same structured protocol involving two of the authors: one person wearing the microphone under evaluation (the wearer, male, 33 years old), and a second person acting as a nearby speaker (the non-wearer, male, 47 years old).



(a) The protocol, with speech from the microphone's wearer following that of the non-wearer, at different distances.



(b) Types of microphones, attachment and placement.

Figure 1: The protocol we followed and the recording equipment we used for running our study.

The protocol consisted of five speaking phases, illustrated in Figure 1a. As shown in the Figure, we start with speech from the non-wearer at varying distances from the microphone's wearer, before switching to speech from the microphone's wearer, at different volumes. During each phase, only one person spoke at a time. The content of the speech was predefined and similar across conditions, with speakers stating their role and the condition this recording was related to. The low-voice condition was included to test whether microphones could reliably capture quieter speech from the wearer without becoming overly sensitive to surrounding voices.

This protocol was intentionally simple and repeatable, and we used it across all microphone configurations to enable direct comparison. By avoiding overlapping speech and turn-taking, we aimed to isolate the effect of microphone placement and distance on speech detection, rather than modeling the full complexity of conversational interaction.

3.1 Microphone Conditions

The first configuration used a **shirt-mounted clip-on microphone** (RODE Wireless GO) attached to the collar of the wearer's shirt. This setup reflects a common choice in laboratory studies and interactive system prototypes. Based on preliminary observations suggesting that the shirt-mounted placement was insufficient to isolate the wearer's speech, we evaluated a second configuration using the same **clip-on microphone mounted on a custom-built**

rigid necklace. This placement positioned the microphone in front of (and closer to) the wearer’s mouth while remaining wearable and hands-free, to mimic recording from a headset. Finally, we evaluated a **commercial lightweight headset** (Sujeetec unidirectional condenser microphone). This represents a more cumbersome apparel than clip-ons but remains reasonably comfortable.

4 Processing Pipeline

All recordings were processed offline. The audio stream was segmented into frames of 30 ms which is the maximum duration supported by the WebRTC VAD Python implementation¹. For each frame, we applied that algorithm, using aggressiveness mode 2 (on a scale from 0 to 3). This mode was selected empirically as a balance between sensitivity to speech and robustness to background noises. We then applied an energy-based filtering step to remove low-energy activations that were unlikely to correspond to intentional speech by the microphone wearer. For each frame classified as speech by the VAD, we computed its root mean square (RMS) amplitude. Figure 2a shows the RMS values over time for 3 recording files replicating the same condition except for the microphone type and placement.

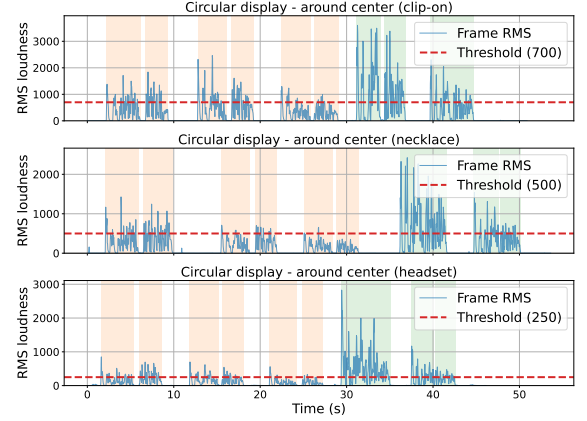
When the RMS value was below threshold (we tried to find suitable values for each condition based on energy levels observed in the corresponding recordings), the VAD output was ignored in the sense that the corresponding frame was classified as “non-speech”. The outcome of this combined process of VAD associated with energy-based filtering produces what we call the *Raw VAD signal*.

Despite these useful measures, VAD decisions are known to be unstable, especially in conversational settings with short pauses, breathing sounds, or overlapping speech. To address this, we additionally applied a temporal smoothing procedure based on a sliding activation window. A window of 30 consecutive frames was used to compute the proportion of frames classified as speech. Window-based speech utterances started once 60% of the current window was classified as speech, and stopped once that was the case for less than 20%. The result of this stage is a *Smoothed VAD signal*, which reflects sustained speech rather than isolated frame-level detections.

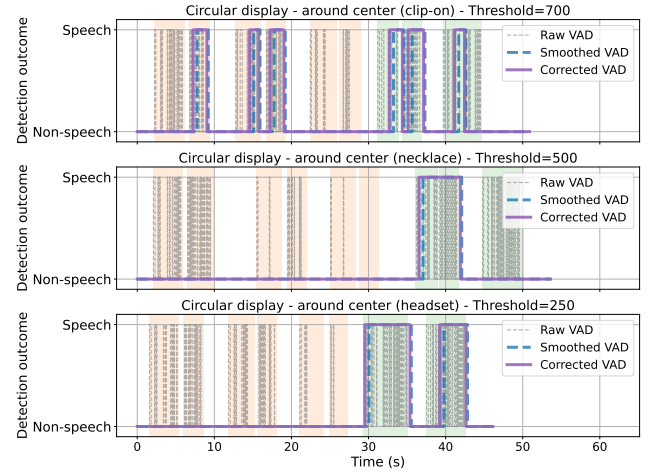
The window-based smoothing introduces a systematic detection delay at speech detection and closure. To compensate for that, we shifted each segment by the expected delay at both its start and end. This correction yields better time alignments that more accurately fit the actual spoken utterances. We thus end up with a *Corrected VAD signal*.

5 Results

The previously mentioned three types of VAD signals are visualized together in Figure 2b to illustrate the purpose and effect of each step but also to facilitate the comparison between microphone configurations and speaker distances. While we can expect any microphone to capture the speech of the non-wearer to a certain extent, the energy levels observed with the clip-on microphone are problematic for making VAD decisions.



(a) Loudness (RMS) values of the recordings over time.



(b) Output of each VAD processing step for the three selected recordings.

Figure 2: Processing results for different microphone configurations. Note that zones with orange and green shades correspond to ground truth annotations of speech utterances, respectively from the non-wearer and the wearer.

We cannot report all metrics due to space constraints but as an example, the median RMS value when the non-wearer was speaking at 1m from the wearer was 435.0 with the shirt-mounted clip-on microphone. Under the same condition, low voice speech from the wearer led to a similar median of 428.0.

This can even be seen visually through Figure 2a, where RMS values for that condition (the upper part of the Figure) are too similar. As seen in Figure 2b, this leads to unreliable VAD decisions with many false positives that the smoothing approach cannot circumvent. For example, 25.21% (at 30cm) and 45.83% (at 1m) of the speech from the non-wearer incorrectly activated the VAD system, while only 60% of the normal speech from the wearer correctly activated it.

The necklace condition improves the situation largely, with clearer differences in energy levels. Having the microphone directly

¹github.com/wiseman/py-webrtcvad

in front of the mouth makes it possible to reliably differentiate normal speech from the wearer from that of the non-wearer, through our simple VAD pipeline (see middle parts in Figure 2).

The commercial headset however produces the best results, with clear differences between wearer and non-wearer normal speech utterances, but with the additional ability to partly capture the wearer's speech at a low volume (differentiating it from non-wearer speech at normal voice level). In terms of quantitative metrics, the median RMS values with the headset were 101.9 and 186.1 for the two sub-parts with speech from the non-wearer 30cm away, while low voice speech from the wearer led to medians of 265.0 and 226.1. With our threshold, this led to 0% of false positives from the non-wearer speech utterances, with 97.37% of the normal speech and even 65.88% of the low voice speech from the wearer being correctly labeled by the VAD system.

When trying to lower the threshold value in order to capture low voice speech from the wearer for the necklace condition as well, we ended up with false positives for speech utterances from the non-wearer at close distance. It is unclear whether the headset's microphone inherently produces better results for our use case (e.g. by being more directional) or whether the microphone's placement on the headset improves over the ad-hoc necklace solution.

6 Design Implications

Our results show that microphone placement plays a decisive role in the reliability of lightweight speaker extraction in shared, collocated settings, which supports RQ2. Even with the same microphone and processing pipelines, a change in placement led to substantial differences in the robustness of speech detection. As we observed, shirt-mounted clip-on microphones are insufficient to isolate a wearer's speech from others in the same room, as they lead to energy levels that are too similar between wearer and non-wearer speech utterances, supporting RQ1. Simple VAD-based pipelines therefore do not suffice to distinguish between them. This challenges an assumption in many user studies that attaching a clip-on microphone to a participant's clothing is "good enough" to associate the recorded speech to that person.

From a design perspective, the three microphone configurations we evaluated reflect a trade-off between convenience, intrusiveness, and robustness. Shirt-mounted clip-on microphones are easy to deploy, unobtrusive, socially accepted, and comfortable, but they offer limited robustness in multi-user environments. Headset microphones (and their emulation through our semi-rigid necklace approach) provide the best target speaker extraction performance, but at the cost of increased visibility, setup effort, and potential participant discomfort. For HCI practitioners, this highlights that microphone choice should be treated as a design decision, based on the study setting and its goals. It should not be considered a simple, purely logistical detail.

Even our simple VAD-based pipeline shows potential for automating speaker identification tasks when associated with the right type of device and placement. With recent and future advances in speech recognition, this opens perspectives for automated analyses that rely on speech attribution (e.g. individual thematic analysis, turn-taking) which are common in multi-user HCI research.

7 Limitations and Future Work

A first limitation is the number of speakers. With only two persons (and one recording run per condition), we do not capture the full spectrum of voices, speaking styles, or morphologies and cannot ensure it generalizes to different profiles. Similarly, our selection of microphone types and placements does not cover different devices, directionality, or clothing material, as we instead focused on common and readily available configurations.

Our recording protocol was highly controlled and avoided overlapping speech. While this was intentional to isolate the effect of placement, natural interactions often involve interruptions and simultaneous speech, which may also raise voice levels. We will therefore extend this evaluation to more natural conversational settings, and have already collected data from real collocated design sessions with 2 to 3 users, each wearing the same commercial headset used in this study. While analysing these sessions falls out of scope of this contribution, this will help further validate the approach.

We should also consider the use of additional information to make more informed decisions on top of VAD and energy levels. This could, for example, involve voice embeddings [12] or visual indicators of speaking activity [2].

Finally, we solely focused on speech activity rather than integrating it with speech recognition or diarization systems. Exploring the impact of microphone choice and placement in complete ASR pipelines is a promising future perspective.

8 Conclusion

Through a controlled evaluation of three microphone placements, we showed that shirt-mounted clip-on microphones are often inadequate for reliable target speaker extraction, while near-mouth and headset placements substantially improve robustness in multi-user contexts. In such settings, researchers should therefore consider using them and weight their benefits against their slightly more intrusive form factor.

While often overlooked, our findings highlight that microphone placement is a critical design variable for shared, speech-based interactive systems and user studies. Researchers and designers can significantly improve the recorded voice input quality by carefully choosing equipment and opting for near-mouth microphones in multi-user contexts. This perspective complements algorithmic research that would push for more complex (AI-based) processing pipelines, as improvements in robustness can be achieved at the physical design level rather than through signal processing alone.

Through empirically grounded guidance on wearable microphone options, this work aims to inform speech-based interaction systems and analysis in HCI research. We hope this pushes future studies to treat audio capture not as a fixed assumption but rather as an integral part of interaction design.

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